Top 10 SQL Interview Questions with Answers

1. What is SQL?

SQL (Structured Query Language) is a standard programming language used to manage and manipulate relational databases. It allows users to retrieve, insert, update, and delete data.

2. What are the different types of SQL commands?

- DDL (Data Definition Language): CREATE, ALTER, DROP
- DML (Data Manipulation Language): INSERT, UPDATE, DELETE
- DQL (Data Query Language): SELECT
- DCL (Data Control Language): GRANT, REVOKE
- TCL (Transaction Control Language): COMMIT, ROLLBACK, SAVEPOINT

3. What is the difference between WHERE and HAVING clause?

- WHERE filters rows before grouping.
- HAVING filters groups after GROUP BY.

Example:

```
SELECT department, COUNT(*) FROM employees
GROUP BY department
HAVING COUNT(*) > 5;
```

4. What are joins in SQL? Name their types.

Joins combine rows from two or more tables based on related columns.

- INNER JOIN
- LEFT JOIN (or LEFT OUTER JOIN)
- RIGHT JOIN (or RIGHT OUTER JOIN)
- FULL JOIN (or FULL OUTER JOIN)
- CROSS JOIN

5. What is a Primary Key?

A Primary Key uniquely identifies each record in a table. It must contain unique values and cannot be NULL.

```
CREATE TABLE students (
student_id INT PRIMARY KEY,
name VARCHAR(50)
);
```

6. What is the difference between DELETE, TRUNCATE, and DROP?

Command	Discription
DELETE	Deletes selected rows, can use WHERE
TRUNCATE	Deletes all rows, faster, no WHERE
DROP	Deletes entire table structure

7. What is a Subquery?

A subquery is a query inside another query. It is used to retrieve data that will be used by the main query.

```
SELECT name
FROM students
WHERE department_id = (
SELECT id FROM departments WHERE name = 'Math'
);
```

8. How do you find duplicate records in a table?

```
SELECT name, COUNT(*)
FROM students
GROUP BY name
HAVING COUNT(*) > 1;
```

9. What is a foreign key?

A foreign key is a column (or combination of columns) in one table that refers to the primary key in another table. It maintains referential integrity.

```
CREATE TABLE orders (
order_id INT,
customer_id INT,
FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);
```

10. How do you find the nth highest salary in SQL?

```
SELECT DISTINCT salary
FROM employees e1
WHERE N - 1 = (
SELECT COUNT(DISTINCT salary)
FROM employees e2
WHERE e2.salary > e1.salary
);
```